

EEHC DISTRIBUTION MATERIALS SPECIFICATION	EDMS 08-400-02
Oil Immersed Coupling Power Transformers With On load Tap-changer And Its Control (ONAN/ONAF Cooling)	1-10-2024

EDMS 08-400-02

SPECIFICATION FOR Oil Immersed Coupling Power Transformers With On load Tap-changer And Its Control (ONAN/ONAF Cooling)

**22/11 Kv (Dyn11) or 11/22Kv (dYN11) from 2.5 MVA up
to 30 MVA**

(Indoor-Outdoor)

Issue: OCT.-2024 / Rev- 2

- This revision contains option items that must be selected by ...EDC before bidding.
- EDMS 25-401 for Mineral Insulating Oil shall be attached.
- EDMS 11-400 for Outdoor Surge Arrester shall be attached if applicable.
- EDMS 26-200 for Neutral Grounding shall be attached.
- EDMS 25-400 for SELF INDICATING ORANGE SILICA GEL CRYSTALS shall be attached.

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1. SCOPE

This specification covers the general requirements of design, manufacture, testing, supply and delivery of **(22/11 KV or 11/22 KV)**, three-phase, two-windings oil immersed (**Step-up or Step-down**) power transformers with on load tap changer and local / remote tap change control panels rated up to **30 MVA** at (ONAN/ONAF) Cooling. The transformers shall be suitable for indoor /outdoor installations of the power networks of Egyptian Electricity Distribution Companies (...EDCs).

2. APPLICABLE STANDARDS

The equipment/material covered in this specification shall comply with the latest edition/amendment of the standards given in Table (1). Where any provision of this specification differs from those of the standards listed below, the provisions of this specification shall apply. Any deviations from the listed standards or the provisions of this specification shall be clearly set out in the offer.

Table (1)

Standard No.	Description
IEC 60076 power transformer	Part 1 General.
	Part2 Temperature rise for liquid immersed transformers.
	Part3 Insulation levels and dielectric tests and external clearance in air.
	Part 5 Ability to withstand short circuit.
	Part7 Loading guide for oil-immersed power transformers.
	Part 10 Determination of sound levels.
IEC60137	Insulated bushings for alternating voltage above 1000V.
IEC60156	Insulating liquids – determination of the breakdown voltage at power frequency – Test method.
IEC60296	Fluids for electrotechnical applications – Unused mineral insulating oils for transformers and switchgears
IEC 60551	Determination of transformer and reactor sound levels.
IEC 60214	Tap Changer

3. SERVICE CONDITIONS

The electrical and mechanical properties of the required transformers should be guaranteed under the environmental conditions given in Table (2).

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Table (2)

Minimum ambient temperature	-5 °C
Maximum ambient temperature	+45 °C (+50 °C as option according to ...EDC)
Maximum relative humidity	95 %
Altitude	Up to 1000 m above sea-level

4. DESIGN CRITERIA

4.1. Rated Power

The transformer should be able to deliver for an unlimited period of time at the main tapping of the tap changer, without exceeding the allowable temperature rise limit, if the applied voltage is equal to the rated voltage and maintained at its rated frequency.

- The maximum continuous power rating is selectable from the following set of standard rates: (2.5MVA-5MVA- 10MVA -15MVA -20MVA-25MVA-30MVA)

4.2. Voltage Ratio

- The ratio of the transformers should be (11KV /22KV) OR (22KV/11KV) according to..... EDC requirement.
- The deviation of the transformation ratio measured at no-load and 50 Hz should not exceed 0.5% from its nominal value.

4.3. Vector Group

The winding connection group of the transformers shall be Delta on Source side and Star Earthed on the Load side in accordance with the vector group (Dyn11 for 22/11k.v) or (dYN 11 for 11/22K.V) as specified in IEC 60076.

4.4. Tapping Range

The transformers should be equipped with 3 phase tap changer:

- For ratings 10 M.V.A to 30 M.V.A., the tap changer is on-load and 9 steps (+2 X 2.5% & -6 X 2.5%) as given in Table (3)
- For ratings 2.5 M.V.A and 5 M.VA, the tap changer is off-circuit and 7 steps (+2 X 2.5% & -4 X 2.5 %)(**as option tap changer is on-load and 9 steps according to ...EDC**).

Table (3)

Tap No	1	2	3	4	5	6	7	8	9
Voltage Regulation%	+5	+2.5	0	-2.5	-5	-7.5	-10	-12.5	-15

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4.5. Overload Capacity

Permissible overload capacity starting from full load 10 % for 2 hr and 20% for 30 min. (without exceeding the permissible heating).

4.6. Short-Circuits Withstand

Transformers should be capable of withstanding, without damage, the thermal and mechanical effects of short circuits at the terminals for 2 sec at least. The short circuit level of 22 kV & 11KV network is 500 MVA

4.7. Temperature Rise limits

The transformer should be able to deliver its max. Continuous ratings, without exceeding the following permissible temperature rise limits: -

• Top oil level temperature rise	45 °C (above ambient temperature)
• Winding temperature rise measured by resistance method	55 °C (above ambient temperature)
• Hot spot	60 °C (above ambient temperature)

4.8. Cooling

Cooling type shall be one of the following options (according to...EDC requirements)

1. ONAN for 100% rated power.
2. ONAN for 100% rated power and ONAF for 120% rated power
3. ONAN for 80% rated power and ONAF for 100% rated power

4.9. Insulation Level

The following test voltages shall be applicable to the transformer: -

– Impulse withstand voltage

(22KV)	125 KV for <u>indoor</u>
	145 KV for <u>outdoor</u>
(11KV)	75 KV for <u>indoor</u>
	95 KV for <u>outdoor</u>

– Power frequency withstand voltage :

50KV/1minute for (22 KV) , 28KV/ 1minute for (11KV)

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4.10. Impedance Voltage

The impedance at mid-tap (normal tap) shall be the following quaternaries at 95°C:

<u>Rated Power</u>	<u>Percentage Impedance at 95°C (Tolerance $\pm 10\%$)</u>
2.5MVA	6%
5MVA	7%
10 MVA	8%
15 MVA	8 %
20MVA	8 %
25MVA	8 %
30MVA	10%

4.11. Bushings

1. The transformer bushings shall be made of porcelain. Generally, it shall comply with the requirements of IEC 60137.
2. The creepage distance should not be less than 2 cm/kV for indoor transformers and 4 cm/kV (4.5 cm/KV optional according to ...EDC requirements) for outdoor transformers.
3. To avoid overvoltage damage each high voltage bushing should be fitted with a duplex arcing horn electrodes made of steel. The vertical distance of arcing horn should be set at (155 mm for 22 K.V/86 mm for 11 K.V) according to VDE0111/261.
4. Neutral bushings should be mounted on the tank cover in a manner such that the minimum phase to phase clearance and phase to earth clearance should not be less than those stated in IEC 60076.
5. The porcelain should not engage directly with the hard metal and gasket should be interposed between them and the surface in contact with the gasket should be unglazed.
6. The neutral bushings and stems should be identical to those provided for phase terminations.
7. Bushing palms should be made of brass and be suitable for the bolting of conductor compression lugs.

4.12. Core

1. The core should be made of low loss steel lamination which are made of cold rolled grain oriented, annealed, high permeability, non-ageing, silicon steel type.
2. The lamination should be insulated with suitable varnish or paint, solid insulation is not acceptable.
3. The core, framework, clamping arrangement & general structure should be mechanically

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robust to withstand any shocks to which they may be subjected during transport, installation, service & faults.

4. The core construction should ensure the efficient cooling of its internal parts, the elimination of core vibration when loaded and the minimization of harmonic voltage especially the third harmonics.
5. The core joints should be properly interleaved, precautions should be taken to keep the noise at the level according to rated capacity (refer to the table of power losses), and at measured distance of one meter, at mid tap & no-load & eliminate vibration in the core when loaded & to diminish harmonic voltage as far as possible especially the third harmonics.

4.13. Windings

1. The winding conductors should be made of the best quality, high conductivity electrolytic copper, complying with the requirements of the IEC publication max.flux density in the yoke should not be more than 1.6 tesla.
2. The windings should be clamped effectively so that they withstand shocks and vibrations during transport and forces produced by the most severe short circuit currents.
3. The primary and secondary windings should be constructed from high conductivity E.C grade copper.
4. All turns of windings should be adequately supported to prevent movement.

4.14. Tank

1. The transformer should be enclosed in a suitable stiffened welded steel tank such that the transformer can be lifted and transported without permanent deformation or oil leakage.
2. The construction should employ weldable mild steel and should be of sufficient strength and rigidity to withstand moving, shipping, handling and effects of S.C. inside the transformer without deformation.
3. Lifting lugs should be provided, suitable for the weight of the transformer, including core and windings, and with the tank filled with oil.
4. Each tank should be provided with jacking lugs suitably positioned for transport. Radiators provided should be detachable fin type and connected to the main tank through flanged valves mounted on the tank at top and bottom and each tank should be fitted with drain valve and air release plug.
5. It must be possible to remove any bushing without removing the tank cover.
6. All nuts and bolts used should be hot dip galvanize and spaced at sufficiently close intervals to avoid buckling of either flange or covers and should provide reasonably uniform compression of the gasket.
7. The transformer tank should be made of best quality plate steel with bolted cover.
8. Pockets should be provided for a stem type thermometer and for the bulbs of temperature indicators where specified. These pockets should be located in the position of maximum

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oil temperature and it must be possible to remove any bulb without lowering the oil level in the tank.

- Captive screwed caps should be provided to prevent the ingress of water to the thermometer pockets when they are not in use.

4.15. Oil expansion conservator

- The transformer should be provided with overhead conservator tank, connected to the main tank as well as the tap changer tank at the highest point to prevent the trapping air or gas under the main tank cover.
- The conservator should be made of sheet steel with an adequate capacity to allow for the extreme expansion of oil level e.g. from ambient temperature to at least 100°C.
- Conservator should have a capacity between highest and lowest indicated oil levels of not less than 7.5% of the total cold oil volume in the transformer and its associated cooling equipment(For rated power >5 M.V.A).
- The conservator is provided with the following:
 - Oil level indicator with two contacts for maximum and minimum levels with sight glass(optional For rated power 2.5M.V.A).
 - silica-gel shall meet the requirements of EDMS 25-400.
 - Oil filling and drain plugs and a connection pipe for connecting the oil to the transformer.

4.16. Transformer Oil

- The oil should meet the requirements of the latest revision of EDMS 25-401.

4.17. Valves

- The transformer should be fitted with necessary valves at the following locations for filling, draining, filtering, conservator isolation etc.

Item	Location	Description
1	Conservator -Main	Filler pipe with cap
2	Conservator-OLTC	Filler pipe with cap
3	Conservator -Main	Drain valve Internal thread with flanged plug. One each end
4	Conservator-OLTC	Drain valve Internal thread with flanged plug.
5	Main tank	Filter Valve
6	Main tank	Drain valve Internal thread with flanged plug

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7	Main tank	A sampling valve for ratings ≥ 5 MVA in case a drain valve without a sampler
8	OLTC	Filter Valve
9	OLTC	Drain Valve
10	Radiator	Isolation Valve (2) per radiator(one at Top and one at Bottom)
11	Radiator	Air bleed plug
12	Radiator	Drain Plug 1 per radiator
13	Conservator -Main	Isolation Valve (For rated power >10 M.V.A).
14	Conservator-OLTC	Isolation Valve(For rated power >10 M.V.A).
15	Buchholz relay	Isolation Valve (For rated power >10 M.V.A).

- All valves should be made of gunmetal having non-rising spindles and should be closed by turning the hand wheel in a clock wise direction.
- All valves should have visual indication of “Open” and “Shut” and should be labeled as to their function.
- All valves which are mounted near the base of the transformer should have mechanical protection provided to prevent damage during transport etc.

4.18. Oil Level Indicator

- Oil level indicator of the magnetic type shall be fitted to the conservator tank to show the oil level at all temperature likely to be experienced in service.
- It shall be marked with normal level at room temperature clearly visible from normal access level.

4.19. Pressure Relief Device

- A pressure relief device of suitable size should be mounted on the tank cover (with a skirt to produce at least 2.5mm in to the tank to prevent gas accumulation) for the rapid release of any pressure that may be generated in the tank.
- The device should be capable of maintaining the oil tightness of the transformer under all conditions of normal service.
- It should prevent the ingress of rain/moisture or oil flow from the transformer after the operation of the device to relieve an internally generated pressure.

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4.20. On Load Tap Changer (OLTC)

1. Normal ratio of transformation of voltages between phases on the primary and secondary windings of the transformer should be 11 KV/22 KV.
2. On-load 9 position Tap Changer and the associated operating mechanism should be able to vary the ratio of transformation in steps of 2.5%.
3. The range of transformation ratio should be +5% to -15%.
4. Voltage trapping should be provided on the high voltage side of the transformer.
5. Insulating oil for the tap changer diverter switch should be separate from the tank oil and fed from a separate compartment in the conservator tank.
6. The tapings should be changed by (Local/Remote) Push button as well as by manual handle. Direction of operation for Raise/Lower tap should be provided. Tap changer mechanism should send signal to initiate Indication/Alarm for the following:
 - a) Tap changer in progress.
 - b) Tap changer incomplete.
 - c) Out of step Indication.
 - d) Supply voltage failure.
 - e) Tap position.
 - f) Tap changer counter.
7. Tap changer operating mechanism motor shall be 3 phase, 380V & 50Hz.
8. The voltage sensing equipment shall prevent operation of the tap-changer if the reference voltage is lost.
9. The relays, switches and push buttons shall be enclosed in a weatherproof cubicle and be readily accessible for maintenance at ground level with the transformer energized.
10. The Motor Drive Unit of the OLTC shall contain a thermostatically controlled anti-condensation heater, a door operated light and a manual handle for emergency and maintenance operation.
11. It shall be weatherproof and dust proof (IP54) and shall be bolted to the transformer main tank in a convenient position such that the operator standing at ground level can carry out maintenance on all equipment contained in the cabinet with the transformer energized.

4.21. Temperature Indicator and Alarm Devices

1. Oil temperature indicator (OTI) and alarm devices of approved design shall be provided to indicate the hottest oil temperature (For rated power >10 M.V.A).
2. For ONAF; winding temperature indicator (WTI) and alarm devices of approved design shall be provided to indicate the hottest spot of windings.
3. The devices shall have a dial type indicator and a pointer to register the highest temperature reached which shall be re-settable.
4. They shall be housed in a lockable marshalling box with ingress protection (IP54).
5. This marshalling box shall be provided of connection of alarms and indications. All

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wiring shall be terminated on standard terminals and labeled with non-ferrous labels.

6. The marshalling box shall contain vertically hinged doors, a thermostatically controlled anti-condensation heater, a door operated light and a general outlet.

4.22. Surge Arrester Mounting Bracket

1. The surge arrester mounting bracket made of steel shall be provided on the transformer main tank and not on the cover plate or fixed to the cover bolts.
2. The bracket shall be hot dip galvanized and suitable to accommodate three nos. of surge arresters on the source side.
3. A surge arrestor should be fixed to the medium voltage bushing. It should meet the requirements of the latest revision of EDMS 11-400.

4.23. Fans (for ONAF cooling only)

1. The power transformer should be supplied with 50Hz fans for proper cooling.
2. No. of fans as per design & transformer rated power .
3. The shape of the built-in fans should be designed to obtain an optimal airflow.
4. The tubes should be made of HDPE (High Density Poly Ethylene) and should be UV resistant, corrosion proof and light-weight.
5. The fans should be produced in accordance with IP54 standard (dust and water resistant) and the materials to be used should meet the standard of insulation class F (resistance to temperatures up to -5°C to 45°C).
6. The fans should have the following features:
 - a. Low level of maintenance and long-life cycle.
 - b. Low ampere draw.
 - c. Built-in thermal protector single/three phase.
 - d. Low noise level.
 - e. Vertical or horizontal airflow.
 - f. Hot dipped Galvanized

4.24. Accessories:

The transformer shall be provided with the following accessories:

1. Buchholz relay with alarm and trip contacts for detection of any internal fault inside the transformer tank.
2. The OLTC should be equipped with a separate tank protected by dependent Buchholz relay.
3. WTI and OTI for transformers (For rated power > 10 M.V.A).
4. Dial type thermometer with two contacts (in case the transformer doesn't contain OTI)
5. Pressure relief device of trip contacts for rated power > 2.5 MVA (optional according to EDC requirements)
6. Drain valve complete with control tap.

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7. Welded lifting shackles.
8. Tank earthing, M 10 stud with nut shall be used.
9. Name plate.
10. Radiator

4.25. Wheels

1. The enclosure base should be fitted with four bi-directional wheels to facilitate moving of the transformers during installation in two perpendicular directions.
2. Wheels should have a locking facility to secure the transformer in its position.

5. TRANSFORMER LOSSES

The transformer should be designed for minimum losses, when comparing between different tenders, the present worth value of the annual capitalized cost of guaranteed losses in the transformers shall be added according to the following formula: -

$$P. w = K \times 8760 \times C [W_i + (LS.F) P^2 W_{cu}]$$

Where:

K= Present worth factor (L.E)

$$K = ((1+i)^N - 1 / i(1+i)^N)$$

I: latest interest rate from Central Bank of Egypt.

N: number of years (20 years).

C= Cost of KWH (According to the max commercial cost of KWH and can be adjusted as per EEHC latest update).

LS.F= Loss factor for the load = 0.4

W_i = Iron (no load) losses in KW at rated voltage and main tapping

W_{cu} = copper (Load) losses in KW at full load and (95°C at main tapping)

P= Peak load (P.U.) = 0.8

- The values of the required losses are given in Table (4)

Table (4)

Power (KVA)	No Load Losses (W)	Load Losses (W) at 95°C	% Z (±10%)	Noise db
2500	3650	22600	6	53
5000	5000	31500	7	53
10000	8500	56000	8	58
15000	13000	62500	8	62
20000	14500	96000	8	68
25000	15000	112000	8	73
30000	20000	130000	10	73

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6. MARKING

- Transformer serial No. and rated power should be engraved on tank cover at LV side.
- Each transformer should be fitted with a name plate of weather proof materials fitted in a visible position, showing the information listed below. Entries on the plate should be indelibly marked by etching or engraving:
 - a) Manufacturers serial number
 - b) Year of manufacture
 - c) IEC Codes
 - d) Customer (.EDC) P.O No.
 - e) Cooling type .
 - f) Type: Conventional /Hermetically Sealed.
 - g) Number of phases
 - h) Rated power
 - i) Rated frequency
 - j) Rated voltages
 - k) Rated currents
 - l) Vector group
 - m) Impedance voltage
 - n) Type of oil
 - o) Total weight/Weight of oil
 - p) Connection diagram
 - q) Short circuit duration
 - r) A table giving tapping voltages of the positions

7. TESTING and INSPECTION

Customer representative has the right to send its representative to attend any or all of these tests. Testing dates shall be advised by the contractor to permit witnessing by Customer representative (s). test certificates shall be sent to customer representative.

7.1. Type Tests:

Type tests should be carried out on one transformer for each type according IEC standard.

The type tests should include and not limited to the following:

1. Full wave lightning impulse voltage withstand test. IEC 60076
2. Temperature rise test. IEC 60076
3. Chemical oil tests.

7.2. Routine Tests:

These tests shall be applied on each transformer in accordance with the requirements of IEC standards for transformers, these shall include the following:

- 1-Measurement of d.c. winding resistance (the allowable deviation in the dc resistance value between phases should be less than 5% at high voltage side and should be less than 10% at low voltage side)
- 2- Turns ratio and polarity check or vector-group symbol at all tap positions. Bushing positions must have permanent markings at this stage of production.
- 3-Measurement of impedance voltage

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- 4-Measurement of load loss.
- 5-Measurement of No-load loss.
- 6-Induced over voltage withstand test.
- 7-Separate – source power frequency voltage withstand test.
- 8-Electrical oil tests.
- 9-Insulation tests between windings and earth and between winding each other.
(absorption factor (R60/R15 > 1.3)
R60: insulation resistance at 60 second.
R15: insulation resistance at 15 second.
- 10- Frequency Response Test.
- 11- Tan (δ) Test.

7.3. Special Tests:

- 1- Measurement of sound level test IEC 60076-10
- 2- Short circuit test. IEC60076 -5
 - The supplier shall submit a short circuit test in an approved lap by the EEHC otherwise the supplier shall comply with the following:
 - a. Short circuit calculations for the coupling transformer need approval shall be submitted to the Extra High Voltage Research Center for review with the same sample coupling transformer with its drawing.
 - b. The supplier shall guarantee the coupling transformer for (3) years, the guarantee remains valid throughout this period or till the short circuit test is submitted whichever comes first.
 - c. The supplier will get approval for only one year and it will be renewed annually if there is no complaint from any EDC during the guarantee period (3 years).

7.4. Penalty and Rejection

7.4.1. Rejection of Material:

If during inspection tests at the works attended by the customer representative engineer any material is found to be defective or not manufactured according to the specifications agreed upon the customer representative engineer have the right to reject such material if he is not convinced that it can satisfactorily be repaired. In such case the contractor shall be obliged to replace the defective material without being entitled to any extra payment whatsoever or to any extension of time of completion of contract.

7.4.2. Penalty and Rejection for Decreased Efficiency

The contractor should keep the No load losses and load losses to be within the guaranteed values, should the loss values, measured from the works tests, exceed the allowable tolerances in the IEC or approved equal standards, then the customer representative shall have the right to reject the transformer with such losses.

However, the measured losses deviate from the guaranteed figures, but lie within the allowable tolerances (10 % total losses), then the excess losses shall be evaluated as per the following formula and the contractor shall be penalized accordingly.

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$$P.W = K \times 8760 \times C [dw_i + (LS.F) (p^2) dw_{cu}] LE$$

Where:

P.W= present worth (L.E) of annual capitalized cost of losses at 11% rate of interest over 20 years.

K= present worth factor (L.E) .

C= cost per KWHR (liable to be adjusted at the time of evaluation of the penalty)

LS.F= Loss factor for the load = 0.4

dw_i = Excess No load losses in KW at normal voltage and main tapping

dw_{cu} = Excess load losses in KW at full load and 95°C at main tapping.

P = peak load in p.u. = 0.8

- No credit shall be given for any decrease in the value of loss(es) below the guaranteed.

7.4.3. Replacement of Rejected Material and Equipment

If according to the above articles, any material or equipment is rejected, the contractor shall be obliged to replace it without extra payment.

8. SUBMITTALS

8.1. Submittals required with tender:

1. The tenderer shall complete and return one legible copy of the attached Data Schedule for each rate of the offered transformers.
2. Dimensional drawings of the offered transformers including interconnection diagrams of all components and assemblies.
3. Original catalogues shall be submitted to facilitate evaluation of the offer.
4. All information should be in English language, clearly readable, with IEC standard symbols and SI units.
5. Type and special test reports/certificates from an independent testing laboratory shall be submitted to __ EDC.
6. Approval certificate issued by Egyptian Electricity Holding Company.
7. List of deviations & clauses to which exception is taken (if any).
8. Spare parts list (option as per ...EDC requirements)
9. The tenderer should submit a list of recommended spare parts.

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8.2. Submittals required following award of contract:

1. The contractor should provide with shipment of the transformer complete installation, operation and maintenance instruction/manuals suitable for the specific supplied item(s), including information from the manufacturer of each purchased accessory.
2. It should include interconnected diagrams, drawings and instructions describing procedures for items that will be shipped to be assembled and interconnected for operation when installed.
3. These should contain strict instructions as required for carrying, moving by skidding or rolling, and lifting the transformer and accessories.
4. The manual should include the instructions for tap changing, enclosure assembly, disassembly, temperature indicator reading, setting procedures and other information designed to assist in proper operation and periodic maintenance of the supplied equipment.
5. Key interlocking should be identified, and serial number should be supplied.
6. All manuals, instructions and supporting drawings and parts lists supplied by the manufacturer and supplemental instruction and data supplied by accessory manufacturers should be in English, clearly readable and arranged in such format to make the subject items easily identifiable.
7. The contractor should provide a complete set of "as manufactured" drawings and wiring diagrams including interconnection diagrams of all components and assemblies of the equipment which assist in operation and maintenance of the equipment.
8. All drawings should be compatible with current industrial standards and good industrial practice.
9. All information should be in English language, clearly readable, with IEC standard symbols, device model, numbers and other requirements.
10. All Dimensions should be in centimeters.

9. GUARANTEE:

- The supplier guarantee the transformer against all defects arising out of faulty design or workmanship, or of defective material for a period of 12 months from putting in service or 18 months from delivery date.
- In case of supplying for the first time, the guarantee period should be extended to 4 years at least from delivery date.

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10. TECHNICAL DATA SCHEDULE

1. The tenderer must fill in thoroughly the attached technical data schedule
2. Any offer does not accompanied with clear and complete technical data schedule shall be rejected.

11. الاشتراطات:

تلتزم الشركة الموردة بتنفيذ الآتي بعد التوريد و قبل إطلاق التيار الكهربائي على المحولات:-

- أ- إرسال أحد المختصين للإشراف علي كل من عملية التركيب (أجزاء المحول – مغير الجهد و لوحة التحكم الخاص به) ، و إختبارات التدشين و الاختبارات الروتينية بالموقع.
- ب- تدريب المهندسين و الفنيين علي عمليات التشغيل و الصيانة

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TECHNICAL DATA SCHEDULE

Power Transformer (11 / 22 KV or 22/11KV)

1- Manufacturer	
2- Type	
3- Standard specification	
4- Rated Voltage (primary/secondary)	
5- Vector group	
6- Rated output at 45°C ambient (MVA)	
7- Method of cooling	
8- Oil type	
9- Material of insulation	
- HV winding	
- LV winding	
- Major insulation	
10- Material of core and its cross-section area (mm ²)	
11- Max. core flux density at full load (Tesla)	
12- Turns ratio	
13- No Load losses (watts)	
14- Load losses at full load current / 95°C winding temperature (Watts)	
15- Resistance per phase at 95°C	For primary (Ohms)
	For secondary (Ohms)
16- Cross-sectional area of cu windings	For primary (mm ²)
	For secondary (mm ²)

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17- Permissible symmetrical S.C current at L.V side terminals for 2sec (KA)	
18- Primary voltage at normal tapping (volts)	
19- Corresponding secondary voltage at	
- No load (volts)	
- Full load at 1 p .f. at 95°C (volts)	
- Full load at 0.8 p.f. at 95°C (volts)	
- Regulation at 1 p.f. at 95°C (%)	
- Regulation at 0.8 p.f .at 95°C (%)	
20- Permissible overload duration without change of specified temp. rise according to IEC 60354	yes [] No []
21- Full load current at LV side (Amps)	
- Full load current at HV side (Amps)	
- No load current at LV side (Amps)	
22- Impedance voltage (%)	
23- Efficiency at normal ratio and 95°C (%)	
- At 100% rated output 1 p.f (%)	
- At 100% rated output 0.8 p.f (%)	
- At 75 % rated output 1 p.f (%)	
- At 75 % rated output 0.8 p.f (%)	
- At 50 % rated output 1.p.f (%)	
- At 50% rated output 0.8 P.F (%)	
- At 25 % rated output 1 p.f (%)	
- At 25 % rated output 0.8 p.f (%)	

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24- Temperature rise at rated output above 45°C ambient temperature	
A- At oil top level (°C)	
B- Winding temperature (°C)	
C- Core temperature (°C)	
25- Tapping switch (on H.V. Side)	
- No. of steps	
- Value of the step (%)	
26- Medium voltage insulators	
- Length of leakage (mm)	
- Arc gap (mm)	
27- Means for firmly bracing of transformer Winding to avoid loosening them due to S.C	
28- Transformers shall be complete with all accessories in tendered specifications [Customer Spec]	yes [] No []
29- Transformers shall be manufactured according to standards mentioned in tendered spec. and will satisfy all requirements in tender specifications	yes [] No []
30- Tender specifications are attached and each page is signed by us	Yes [] No []
31- Overall dimensions (Length x Width x Height)(mm x mm x mm)	
32- Total weight of trans. With oil (kgm)	
Weight of oil (kgm)	
33- Noise level (db)	
34- Colour of silica gel	

I/we guarantee the correctness of above data for transformers we are offering to Customer Company for Distributing Electricity.

Date: / /

Signature of Tenderer: -----